

Mini Project Report under Python Programming Lab

HOTEL MANAGEMENT SYSTEM

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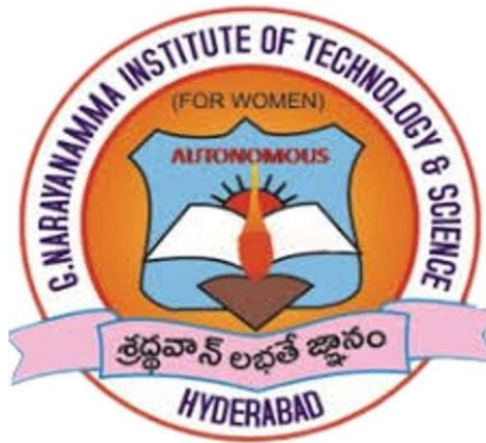
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ABSTRACT

The Hotel Management System is a GUI-based desktop application developed using Python and the Tkinter library. The main objective of this system is to automate hotel operations such as room booking, customer information management, billing, food ordering, and checkout processes.

The application provides an easy-to-use graphical interface that enables hotel staff to efficiently manage daily hotel activities. The system maintains room availability, calculates room charges based on the number of days stayed, supports multiple payment modes, and provides restaurant services.

This project eliminates manual work, reduces errors, saves time, and improves operational efficiency. It is especially useful for small and medium-sized hotels and is also ideal for academic learning purposes.

INTRODUCTION

Hotel management involves various tasks such as maintaining room records, handling customer bookings, processing payments, and managing food services. Traditionally, these operations are performed manually, which can lead to errors, inefficiency, and poor customer service.

With the advancement of technology, software-based solutions have become essential. This project focuses on developing a Hotel Management System using Python Tkinter, which provides a simple yet effective solution for managing hotel operations.

The application uses Python dictionaries for data storage and Tkinter widgets for the graphical interface. The system ensures real-time updates of room availability and provides interactive dialogs for user input and confirmation.

LITERATURE SURVEY

Hotel management systems have evolved significantly from traditional manual record-keeping methods to computerized and automated software solutions. Earlier, hotels relied on paper registers and manual billing systems to record customer details, room bookings, and payments. These methods were time-consuming, prone to human errors, and made it difficult to track room availability and generate accurate bills. Maintaining historical records and retrieving data also posed major challenges in such manual systems.

With the introduction of computers, spreadsheet-based and basic desktop applications were adopted to improve efficiency. However, these systems still required significant manual intervention and lacked user-friendly interfaces and real-time updates. Modern hotel management software focuses on graphical user interfaces (GUI), automation, and modular design to simplify hotel operations. GUI-based systems reduce complexity, improve accuracy in billing, and provide better interaction between users and the system.

Recent studies and applications highlight the use of high-level programming languages like Python along with GUI frameworks such as Tkinter for developing lightweight hotel management systems. These systems are cost-effective, easy to maintain, and suitable for small and medium-scale hotels. By automating processes like room allocation, payment calculation, food ordering, and checkout, modern hotel management systems enhance operational efficiency and customer satisfaction. The proposed system follows these principles by implementing a GUI-based, automated solution using Python and Tkinter.

PROBLEM STATEMENT

To design and implement a GUI-based Hotel Management System that automates:

Customer registration

Room allocation and availability tracking

Payment processing

Food ordering services

Checkout operations

The system must be user-friendly, reliable, and executable only in environments with graphical display support.

OBJECTIVES

To develop a user-friendly hotel management application

To automate room booking and checkout

To maintain customer details efficiently

To calculate room and food bills automatically

To reduce manual effort and errors

To provide multiple payment options

SYSTEM REQUIREMENTS

Hardware Requirements

Computer/Laptop

Minimum 4 GB RAM

Keyboard and Mouse

Software Requirements

Operating System: Windows / Linux

Python 3.x

Tkinter library

IDE: VS Code / PyCharm / IDLE

SYSTEM DESIGN

Data Structures Used

Dictionaries for room details

Dictionaries for restaurant menu

Global variables for booking details

GUI Design

Tkinter widgets such as:

Labels

Buttons

Entry fields

Radio buttons

Message boxes

Dialog boxes

MODULES

Room Management Module

This module stores information about hotel rooms such as room number, type (AC/Non-AC), capacity, availability, and rate per day.

It updates room availability automatically during booking and checkout.

Customer Details Module

Collects customer information such as:

Name

Phone number

Address

Aadhar number

Input validation ensures that no field is left empty.

Booking Module

Allows customers to:

Select an available room

Enter the number of days of stay

The module calculates the total room charge automatically.

Payment Module

Supports multiple payment methods:

PhonePe

Card

Cash

After successful payment, the room is allocated and booking count is updated.

Food Service Module

Displays a restaurant menu and allows:

Room food ordering

Table booking

The total food bill is calculated and payment is processed.

Checkout Module

Handles:

Customer checkout

Extra day charge calculation

Room availability reset after checkout

IMPLEMENTATION

The Hotel Management System is implemented using the Python programming language with the Tkinter library, which provides tools for building graphical user interface (GUI) applications. The application follows an event-driven programming approach, where user actions such as button clicks, data entry, and selections trigger specific functions. This ensures smooth interaction between the user and the system. Before launching the GUI, the program verifies the availability of a graphical display environment to prevent execution errors in non-GUI platforms, such as online compilers.

Data related to rooms, customer bookings, and restaurant menu items are managed using Python dictionaries, allowing quick access, easy updates, and efficient data handling without the need for an external database. The interface is designed using Tkinter widgets such as labels, entry fields, buttons, radio buttons, and dialog boxes. Each functional screen (customer details, room selection, payment, food service, and checkout) is implemented as a separate function. The window is refreshed dynamically by clearing previous widgets, ensuring seamless navigation between different modules. Automated calculations are used for room charges, food bills, and extra stay charges, thereby reducing human errors and improving accuracy. Overall, the implementation emphasizes simplicity, modularity, and usability, making the system reliable and easy to extend in the future.

IMPORTANT CODE SNIPPETS

Checking GUI Environment

```
if os.name != "nt":  
  
    if "DISPLAY" not in os.environ:  
  
        sys.exit(1)
```

Room Data Storage

```
rooms = {  
  
    101: {"type": "AC", "capacity": 2, "available": True, "rate": 1500}  
  
}
```

Room Charge Calculation

```
total_amount = current_booking["days"] * current_booking["rate"]
```

Updating Room Availability

```
rooms[room_no]["available"] = False
```

Food Bill Calculation

```
total = sum(menu_items[item] for item in selected_item)
```

ADVANTAGES

Easy to use

Time-saving

Reduces manual workload

Accurate billing

Cost-effective solution

LIMITATIONS

No permanent database storage

Single-user system

Data is lost after application closes

Not suitable for large hotels

FUTURE ENHANCEMENTS

Database integration (MySQL/SQLite)

Multi-user login system

Online booking facility

Invoice and bill generation

Report generation

Cloud-based system

RESULTS

The Hotel Management System developed using Python and Tkinter produced effective and reliable results during implementation and testing, demonstrating its capability to automate key hotel operations. The system successfully handled customer registration by accurately capturing essential details such as name, phone number, address, and Aadhar number, while ensuring proper input validation to avoid incomplete or incorrect entries. This helped maintain data consistency and improved the reliability of booking records. The room management functionality worked efficiently by displaying only available rooms to the user and updating room availability immediately after a successful booking. This real-time update mechanism prevented double booking of rooms and ensured smooth allocation of resources.

The booking and payment module delivered accurate and transparent billing by automatically calculating room charges based on the number of days selected and the predefined room rate. Multiple payment modes such as PhonePe, card, and cash were integrated seamlessly, providing flexibility and ease of use for customers and hotel staff. Upon successful payment, confirmation messages were displayed clearly, improving user confidence in the system. The system also maintained a running count of total bookings, which provided basic operational insight and helped track hotel occupancy trends.

The food service module yielded positive results by offering an interactive restaurant menu where users could select multiple items. The application accurately calculated the total food bill based on selected menu items, eliminating manual calculation errors. The separate payment process for food services worked smoothly and enhanced the overall customer experience by integrating restaurant services into the hotel management workflow.

The checkout module functioned reliably by validating room numbers and ensuring that only occupied rooms could be checked out. It accurately calculated additional charges when customers exceeded their booked stay duration and informed users about extra payments clearly. After checkout, room availability was reset automatically, allowing the room to be reused for future bookings. Throughout the application, the graphical user interface provided clear navigation, responsive controls, and informative message dialogs, making the system easy to operate even for users with minimal technical knowledge.

Overall, the results indicate that the Hotel Management System significantly improves efficiency, accuracy, and ease of operation compared to manual methods. The system reduces human errors, saves time, and ensures better management of hotel resources. These outcomes confirm that the application is well-suited for small-scale hotels and academic projects and provides a strong foundation for future enhancements such as database integration and multi-user support.

CONCLUSION

The Hotel Management System developed using Python Tkinter successfully automates essential hotel operations such as room booking, customer management, billing, food ordering, and checkout. The project demonstrates effective use of GUI programming and modular design. It is suitable for academic purposes and can be extended into a full-scale hotel management application in the future.

